

2021-Ph-H1-Q8

Section: Our Dynamic Universe

Topic: Collisions, Explosions and Impulse

Which graph shows the largest impulse?

Solution:

Impulse = **area under force–time graph.**

Compare areas: height $\times$ width.

Answer: E

Guidance for Students:

Impulse = change in momentum = Ft .

The graph with the largest area under it gives the largest impulse.

Revision Tip:

Area under F – t graph = impulse = change in momentum.

Units: Ns or $\text{kg}\cdot\text{m/s}$.